BuildCore SQL + Project Verification Guide for Presentation

SQL commands and live checks for demonstrating BuildCore

This document contains the most important SQL commands and verification checks that demonstrate the project clearly during a presentation.

1. Start the project

Open PowerShell in the project folder and run:

POWERSHELL

```
cd "c:\Users\ishan\OneDrive\Desktop\College_Files\SEM-5\Database Systems\BuildCore_DB_Project"
.\.venv\Scripts\Activate.ps1
flask run --debug
```

Then open:

TEXT

```
http://127.0.0.1:5000
```

Login:

TEXT

```
Use the demo administrator account configured for your local installation. Avoid sharing real credentials in
  a presentation copy.
```

2. Connect to PostgreSQL

Use the database name from the project configuration:

POWERSHELL

```
psql -U postgres -h 127.0.0.1 -p 5432 -d buildcore
```

When prompted, enter the PostgreSQL password configured for your local installation. Do not include a real database password in a shared document.

Useful PostgreSQL commands:

SQL

```
\l
\dt
\d projects
\d project_tasks
\d material_usage
\d expenses
```

These commands show that the database and project tables exist.

3. Quick database verification queries

3.1 Show the main tables and row counts

SQL

```
SELECT
    (SELECT COUNT(*) FROM projects) AS total_projects,
    (SELECT COUNT(*) FROM workers) AS total_workers,
    (SELECT COUNT(*) FROM materials) AS total_materials,
    (SELECT COUNT(*) FROM equipment) AS total_equipment,
    (SELECT COUNT(*) FROM suppliers) AS total_suppliers,
    (SELECT COUNT(*) FROM project_tasks) AS total_tasks,
    (SELECT COUNT(*) FROM purchases) AS total_purchases,
    (SELECT COUNT(*) FROM expenses) AS total_expenses;
```

This is a strong presentation query because it immediately proves that the system has real data.

3.2 Show all projects

SQL

```
SELECT project_id, project_name, project_type, client_name, location, status, budget
FROM projects
ORDER BY project_id;
```

3.3 Show all workers

SQL

```
SELECT worker_id, name, role, skill, status, daily_rate
FROM workers
ORDER BY worker_id;
```

3.4 Show materials in stock / material catalog

SQL

```
SELECT material_id, material_name, unit, unit_cost
FROM materials
ORDER BY material_id;
```

3.5 Show suppliers

SQL

```
SELECT supplier_id, supplier_name, contact_person, phone, email
FROM suppliers
ORDER BY supplier_id;
```

4. Project management verification queries

4.1 Show assigned project tasks

SQL

```
SELECT p.project_name, tt.task_name, pt.status, pt.completion_percentage, pt.due_date
FROM project_tasks pt
JOIN projects p ON pt.project_id = p.project_id
JOIN task_types tt ON pt.task_type_id = tt.task_type_id
ORDER BY p.project_name, pt.due_date;
```

This demonstrates task planning and progress tracking.

4.2 Show task completion progress per project

SQL

```
SELECT project_name, progress_percentage, total_tasks, completed_tasks
FROM project_progress_view
ORDER BY progress_percentage DESC;
```

This is a very nice query for presentation because it shows the project progress view in action.

4.3 Show workers assigned to projects

SQL

```
SELECT p.project_name, w.name, pw.role_on_project, pw.assigned_date, pw.release_date
FROM project_workers pw
JOIN projects p ON pw.project_id = p.project_id
JOIN workers w ON pw.worker_id = w.worker_id
ORDER BY p.project_name, w.name;
```

4.4 Show workers assigned to specific tasks

SQL

```
SELECT p.project_name, tt.task_name, w.name, tw.assigned_hours
FROM task_workers tw
JOIN project_tasks pt ON tw.project_task_id = pt.project_task_id
JOIN projects p ON pt.project_id = p.project_id
JOIN task_types tt ON pt.task_type_id = tt.task_type_id
JOIN workers w ON tw.worker_id = w.worker_id
ORDER BY p.project_name, tt.task_name;
```

5. Material and equipment management verification

5.1 Show material usage across projects

SQL

```
SELECT p.project_name, m.material_name, mu.quantity_allocated, mu.quantity_used
FROM material_usage mu
JOIN projects p ON mu.project_id = p.project_id
JOIN materials m ON mu.material_id = m.material_id
ORDER BY p.project_name, m.material_name;
```

5.2 Show equipment assignments

SQL

```
SELECT p.project_name, e.equipment_name, e.equipment_type, ea.start_date, ea.end_date, ea.hours_used
FROM equipment_assignments ea
JOIN project_tasks pt ON ea.project_task_id = pt.project_task_id
JOIN projects p ON pt.project_id = p.project_id
JOIN equipment e ON ea.equipment_id = e.equipment_id
ORDER BY p.project_name, ea.start_date;
```

5.3 Show dynamic equipment status

SQL

```
SELECT equipment_name, equipment_type, computed_status, hourly_rate
FROM equipment_status_view
ORDER BY equipment_name;
```

5.4 Show dynamic worker status

SQL

```
SELECT name, role, skill, computed_status, daily_rate
FROM worker_status_view
ORDER BY name;
```

6. Financial verification queries

6.1 Show all purchases and status

SQL

```
SELECT p.purchase_id, s.supplier_name, pr.project_name, p.purchase_date, p.status
FROM purchases p
JOIN suppliers s ON p.supplier_id = s.supplier_id
JOIN projects pr ON p.project_id = pr.project_id
ORDER BY p.purchase_date DESC;
```

6.2 Show total purchase amount per order

SQL

```
SELECT p.purchase_id, s.supplier_name, pr.project_name, pv.total_amount, p.status
FROM purchases_with_total_view pv
JOIN purchases p ON pv.purchase_id = p.purchase_id
JOIN suppliers s ON p.supplier_id = s.supplier_id
JOIN projects pr ON p.project_id = pr.project_id
ORDER BY pv.total_amount DESC;
```

6.3 Show unified financial ledger

SQL

```
SELECT project_id, category, description, amount, transaction_date
FROM expense_ledger_view
ORDER BY transaction_date DESC;
```

6.4 Show total project spending

SQL

```
SELECT project_id,
       SUM(amount) AS total_spend
FROM expense_ledger_view
GROUP BY project_id
ORDER BY total_spend DESC;
```

7. Best presentation SQL queries to show the project

If you want only the strongest 5 queries for live demo, use these:

SQL

```
SELECT * FROM projects;
SELECT * FROM project_progress_view;
SELECT * FROM project_workers;
SELECT * FROM material_usage;
SELECT * FROM expense_ledger_view;
```

These queries clearly show that the project covers:

- project planning
- task tracking
- workforce allocation
- material control
- financial tracking

8. What to say while presenting

You can explain:

> *“This system is designed for construction project management. The database stores project information, team assignments, task progress, materials, equipment, suppliers, and financial transactions. The application uses relational tables to link projects with tasks, workers, purchases, and expenses. The SQL queries prove that data is being recorded and monitored in real time.”*

9. Verification checklist

Before presenting, confirm all of these:

[ ] PostgreSQL is running

[ ] Database buildcore exists

[ ] Tables are created

[ ] Projects have data

[ ] Workers and tasks are assigned

[ ] Material usage is stored

[ ] Purchase and expense records exist

[ ] Flask app is running successfully

[ ] Login works with the locally configured demo administrator account

10. Final demo summary

A strong final demonstration flow is:

1. Start Flask app
2. Open the app in browser
3. Login as admin
4. Show project dashboard
5. Run SQL queries in psql
6. Show project data, task progress, materials, and financial ledger
7. Explain how the relational database supports the application

This combination of web app + SQL verification makes the project presentation clear and convincing.